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Display device and mobile terminal

Abstract

A display device and a mobile terminal are provided in embodiments of the present application. A display portion of a display panel in the display device provided in the embodiments includes a plane display portion and a first arc surface display portion connected to the plane display portion and a bending portion. In a direction of a light-emitting side of a display module, a height of a horizontal surface where a top surface of an encapsulation structure is located is greater than or equal to a height of a horizontal surface where a top surface of a front frame is located, so as to eliminate occurrence of motion interference caused by a touch between fingers and an edge of the front frame when swiping a screen in prior art.

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Background/Summary

BACKGROUND OF DISCLOSURE

(1) This application is a National Phase of PCT Patent Application No. PCT/CN2022/103060 having International filing date of Jun. 30, 2022, which claims the benefit of priority of Chinese Patent Application No. 202210646155.7, filed Jun. 8, 2022, the contents of which are all incorporated herein by reference in their entirety.

Technical Field

(2) The present application relates to a technical field of displays, and specifically, to a display device and a mobile terminal.

Description of Prior Art

(3) With development of electronic technology, electronic devices such as mobile phones and tablet computers have become an indispensable part of people's daily life and work. Users have higher and higher requirements for appearance design of display panels.

(4) In existing display panels, in order to improve a screen-to-body ratio and reduce a size of a frame, a pad bending technology is usually used, which makes a protruding end of a flexible polyimide (PI) substrate of the display panel bend to a back of a display, so as to realize a narrow frame of a non-display area. However, in a process of matching the existing display panel with a front frame, due to existence of a bending portion, the front frame must be set with a reserved space, and the bending portion is accommodated in the reserved space. Therefore, a height of a horizontal plane where a top surface of the front frame is located is usually set to be greater (about 2 mm) than a height of a horizontal plane where a top surface of a display portion is located. When users swipe the display device, fingers will touch an edge of the front frame, resulting in adverse effects such as occurrence of motion interference.

BRIEF SUMMARY OF DISCLOSURE

Technical Problem

(5) A display device provided in embodiments of the present application is used for alleviating deficiencies in related technologies.

Technical Schemes

(6) In order to realize above-mentioned functions, a technical scheme provided in the embodiments of the present application is as follows.

(7) A display device is provided in the embodiment of the present application, which comprises a front frame and a display module located in the front frame, and the display module comprises: a display panel comprising a display portion and a binding portion arranged opposite to each other, and a bending portion connected between the display portion and the binding portion; and the display portion comprising a plane display portion and a first arc surface display portion connected between the plane display portion and the bending portion; an encapsulation structure arranged on a light-emitting side of the display panel; and a supporting layer located on one side of the display portion close to the binding portion, and the supporting layer comprising a first supporting sub portion arranged corresponding to the plane display portion and a second supporting sub portion arranged corresponding to the first arc surface display portion; and a shape of the second supporting sub portion being matched with a shape of the first arc surface display portion; wherein in a direction of a light-emitting side of the display module, a height of a horizontal plane where a top surface of the encapsulation structure is located is greater than or equal to a height of a horizontal plane where a top surface of the front frame is located.

(8) In the display device provided in an embodiment of the present application, the front frame comprises a bottom frame located on one side of the supporting layer away from the display panel and a first frame located on one side of the bottom frame close to the first arc surface display portion, and the first frame is perpendicular to the bottom frame; and in a direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the first frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

(9) In the display device provided in an embodiment of the present application, the display portion comprises a second arc surface display portion extending along a direction of the plane display portion away from the first arc surface display portion; the supporting layer comprises a third supporting sub portion extending along a direction of the first supporting sub portion away from the second supporting sub portion; the third supporting sub portion is arranged opposite to the second arc surface display portion, and a shape of the third supporting sub portion is matched with a shape of the second arc surface display portion; the front frame further comprises a second frame located on one side of the bottom frame close to the second arc surface display portion, and the second frame is perpendicular to the bottom frame; and wherein in the direction perpendicular to

the display panel, a height of a horizontal surface where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

(10) In the display device provided in an embodiment of the present application, a radius of curvature of the first arc surface display portion is same as a radius of curvature of the second arc surface display portion.

(11) In the display device provided in an embodiment of the present application, the display device further comprises a binding component connected to the binding portion, and in a direction from the second frame pointing to the first frame, a distance between the binding portion and the bottom frame gradually decreases.

(12) In the display device provided in an embodiment of the present application, the display device comprises a supporting member, and the supporting member is located between the binding portion and the first supporting sub portion; and wherein in the direction from the second frame pointing to the first frame, a thickness of the supporting member gradually increases.

(13) In the display device provided in an embodiment of the present application, the display module comprises at least two non-bending areas and a bendable area arranged between adjacent two of the non-bending areas; at least a part of the plane display portion is located in the bendable area, and both the first arc surface display portion and the second arc surface display portion are located in the non-bending areas; and wherein the first supporting sub portion comprises a grid structure located in the bendable area.

(14) In the display device provided in an embodiment of the present application, a material of the supporting layer is metal.

(15) In the display device provided in an embodiment of the present application, the display portion comprises a rolling portion and a non-rolling portion connected between the rolling portion and the bending portion; the display device comprises a rolling shaft located on the one side of the supporting layer away from the display panel and arranged close to the rolling portion; and the rolling portion is wound on the rolling shaft, and the rolling shaft is used to roll up or unroll the display module; and wherein the non-rolling portion comprises the first arc surface display portion, and the rolling portion comprises a second arc surface display portion along a direction of the plane display portion away from the first arc surface display portion; when the display panel is in a unrolled state, in the direction perpendicular to the display panel, the second arc surface display portion and the first arc surface display portion are symmetrically arranged with respect to a horizontal centerline of the plane display portion.

(16) In the display device provided in an embodiment of the present application, the first supporting sub portion comprises a supporting rolling portion arranged corresponding to the rolling portion, and a shape of the supporting rolling portion is matched with a shape of the rolling portion; wherein the supporting rolling portion is wound on the rolling shaft, and the supporting rolling portion comprises a plurality of hollowing portions; and wherein the front frame further comprises a second frame located on one side of the bottom frame close to the rolling portion, and the second frame is perpendicular to the bottom frame; and in the direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

(17) In the display device provided in an embodiment of the present application, the display device further comprises a binding component connected to the binding portion and a supporting member located between the binding portion and the first supporting sub portion; and wherein in a direction from the second frame pointing to the first frame, a thickness of the supporting member gradually increases.

(18) In the display device provided in an embodiment of the present application, the encapsulation structure comprises a flexible substrate and a flexible cover plate arranged in layers, and the

flexible substrate is located between the display panel and the flexible cover plate; and wherein a material of the flexible substrate is one of glass cover plate and ultra-thin glass; and the flexible cover plate is a combination of a cover plate material protective window and an optically clear adhesive, and a material of the cover plate material protective window is one of polyethylene glycol terephthalate and colorless polyimide.

(19) In the display device provided in an embodiment of the present application, the mobile terminal comprises a terminal body and a display device, the terminal body and the display device are integrated, and the display device comprises a front frame and a display module located in the front frame, and the display module comprising: a display panel, the display panel comprising a display portion and a binding portion arranged opposite to each other, and a bending portion connected between the display portion and the binding portion; and the display portion comprising a plane display portion and a first arc surface display portion connected between the plane display portion and the bending portion; an encapsulation structure arranged on a light-emitting side of the display panel; and a supporting layer located on one side of the display portion close to the binding portion, and the supporting layer comprising a first supporting sub portion arranged corresponding to the plane display portion and a second supporting sub portion arranged corresponding to the first arc surface display portion; and a shape of the second supporting sub portion being matched with a shape of the first arc surface display portion; wherein in a direction of the light-emitting side of the display module, a height of a horizontal plane where a top surface of the encapsulation structure is located is greater than or equal to a height of a horizontal plane where a top surface of the front frame is located.

(20) In the display device provided in an embodiment of the present application, the front frame comprises a bottom frame located on one side of the supporting layer away from the display panel and a first frame located on one side of the bottom frame close to the first arc surface display portion, and the first frame is perpendicular to the bottom frame; and in a direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the first frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

(21) In the display device provided in an embodiment of the present application, the display portion comprises a second arc surface display portion extending along a direction of the plane display portion away from the first arc surface display portion; the supporting layer comprises a third supporting sub portion extending along a direction of the first supporting sub portion away from the second supporting sub portion; the third supporting sub portion is arranged opposite to the second arc surface display portion, and a shape of the third supporting sub portion is matched with a shape of the second arc surface display portion; the front frame further comprises a second frame located on one side of the bottom frame close to the second arc surface display portion, and the second frame is perpendicular to the bottom frame; and wherein in the direction perpendicular to the display panel, a height of a horizontal surface where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

(22) In the display device provided in an embodiment of the present application, a radius of curvature of the first arc surface display portion is same as a radius of curvature of the second arc surface display portion.

(23) In the display device provided in an embodiment of the present application, the display device further comprises a binding component connected to the binding portion, and in a direction from the second frame pointing to the first frame, a distance between the binding portion and the bottom frame gradually decreases.

(24) In the display device provided in an embodiment of the present application, the display device comprises a supporting member, and the supporting member is located between the binding portion and the first supporting sub portion; and wherein in the direction from the second frame pointing to

the first frame, a thickness of the supporting member gradually increases.

(25) In the display device provided in an embodiment of the present application, the display module comprises at least two non-bending areas and a bendable area arranged between adjacent two of the non-bending areas; at least a part of the plane display portion is located in the bendable area, and both the first arc surface display portion and the second arc surface display portion are located in the non-bending areas; and wherein the first supporting sub portion comprises a grid structure located in the bendable area.

(26) In the display device provided in an embodiment of the present application, a material of the supporting layer is metal.

Advantageous Effect

(27) The display device and the mobile terminal are provided in the embodiments of the present application. The display device comprises the front frame and the display module located in the front frame. The display module comprises the supporting layer, the display panel, and the encapsulation structure arranged in layers. The display portion is set to comprise the plane display portion and the first arc surface display portion connected between the plane display portion and the bending portion, which can reduce the height of the front frame of the display device. In a direction of the light-emitting side of the display module, the height of the horizontal plane where the top surface of the encapsulation structure is located is greater than or equal to the height of the horizontal plane where the top surface of the front frame is located, so as to eliminate occurrence of adverse phenomena such as motion interference caused by a touch between fingers and an edge of the front frame when swiping a screen in the prior art. In addition, the supporting layer is set to comprise the first supporting sub portion arranged corresponding to the plane display portion and the second supporting sub portion arranged corresponding to the first arc surface display portion, and the shape of the second supporting sub portion is matched with the shape of the first arc surface display portion, so as to support the display panel.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The technical scheme and other advantageous effects of the present application will be apparent through the detailed description of a specific implementation mode of the present application in combination with the accompanying drawings.

(2) FIG. 1 is a schematic structural diagram of a display device in prior art.

(3) FIG. 2 is a first schematic structural diagram of a display device provided in an embodiment of the present application.

(4) FIG. 3 is a second schematic structural diagram of the display device provided in an embodiment of the present application.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(5) A display panel and a display device are provided in the embodiment of the present application. In order to make a purpose, a technical scheme, and an effect of the present application clearer and more accurate, the following is a further detailed description of the present application with reference to the attached drawings and examples. It should be understood that the specific embodiments described here are only used to explain the present application and are not used to limit the present application.

(6) Please refer to FIG. 2 and FIG. 3, the display device **1** is provided in the embodiment of the present application and includes a front frame **20** and a display module **10** located in the front frame **20**. The display module **10** includes: a display panel **100**, the display panel **100** includes a display portion **110** and a binding portion **130** arranged opposite to each other, and a bending portion **120** connected between the display portion **110** and the binding portion **130**. The display

portion **110** includes a plane display portion **111** and a first arc surface display portion **112** connected between the plane display portion **111** and the bending portion **120**.

(7) An encapsulation structure **700** is arranged on a light-emitting side of the display panel **100**.

(8) A supporting layer **200** is located on one side of the display portion **110** close to the binding portion **130**. The supporting layer **200** includes a first supporting sub portion **210** arranged corresponding to the plane display portion **111**, and a second supporting sub portion **220** arranged corresponding to the first arc surface display portion **112**. A shape of the second supporting sub portion **220** is matched with a shape of the first arc surface display portion **112**.

(9) Wherein in a direction of the light-emitting side of the display module **10**, a height of a horizontal plane where a top surface of the encapsulation structure **700** is located is greater than or equal to a height of a horizontal plane where a top surface of the front frame **20** is located.

(10) It should be noted that at present, with development of electronic technology, electronic devices such as mobile phones and tablet computers have become an indispensable part of people's daily life and work. Users have higher and higher requirements for an appearance design of the display panel **100**. In organic light emitting diode (OLED) technology, a pad bending technology is usually used to improve a screen-to-body ratio and reduce a size of a frame. As shown in FIG. 1, it is a schematic structural diagram of a display device in the prior art. Wherein the display device **1** includes a front frame **20** and a display module **10** located in the front frame **20**. The display module **10** includes a supporting layer **200**, a heat dissipation layer **400**, a display panel **100**, a functional film layer **600**, and an encapsulation structure **700** arranged in layers. Wherein the display panel **100** includes a display area A, a binding area C away from one side of the display area A, and a bending area B located between the display area A and the binding area C. The display panel **100** includes a flat display portion **110'** located in the display area A, a bending portion **120** located in the bending area B, and a binding portion **130** located in the binding area C and arranged opposite to the flat display portion **110'**.

(11) The display device **1** further includes a first backplane **310** located between the flat display portion **110'** and the supporting layer **200**, and a second backplane **320** and a supporting member **500** both located between the binding portion **130** and the supporting layer **200**. Wherein due to existence of the bending portion **120**, the front frame **20** must be provided with a reserved space (not shown in figure), and the bending portion **120** is accommodated in the reserved space. In a direction of a light-emitting side of the display module **10**, a height of a horizontal plane where a top surface of the front frame **20** is located is greater (about 2 mm) than a height of a horizontal plane where a top surface of the display portion **110** is located, so as to prevent fine particles in external environment from entering the display module **10** during use of the display device **1**, which will affect service life of the display module **10**. However, it is inevitable that when the user swipes the display device **1**, fingers will touch an edge of the front frame **20**, resulting in adverse effects such as motion interference.

(12) It can be understood that the embodiment of the present application reduces the height of the front frame **20** in the existing display device **1** by setting the display portion **110** to include the plane display portion **111** and the first arc surface display portion **112** connected between the plane display portion **111** and the bending portion **120**. In the direction of the light-emitting side of the display module **10**, the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is greater than or equal to the height of the horizontal plane where the top surface of the front frame **20** is located, so as to eliminate the occurrence of adverse phenomena such as the motion interference caused by fingers touching the edge of the front frame **20** when swiping the screen in the prior art. In addition, the supporting layer **200** includes the first supporting sub portion **210** corresponding to the plane display portion **111** and the second supporting sub portion **220** corresponding to the first arc surface display portion **112**. The shape of the second supporting sub portion **220** is matched with the shape of the first arc surface display portion **112**, so as to support the display panel **100**.

(13) The technical scheme of the present application is described in combination with specific embodiments.

(14) In an embodiment, please refer to FIG. 2, which is a first schematic structural diagram of the display device provided in the embodiment.

(15) The display device **1** is provided in the embodiment, and the display device **1** includes the front frame **20** and the display module **10** located in the front frame **20**. The display module **10** includes the display panel **100**, a functional film layer **600** arranged on the light-emitting side of the display panel **100**, the encapsulation structure **700** located on one side of the functional film layer **600** away from the display panel **100**, and the supporting layer **200** arranged on a backlight side of the display panel **100**. It should be noted that an organic light emitting diode (OLED) display panel **100** is taken as an example to describe the technical scheme of the present application in the embodiment.

(16) In the embodiment, the display panel **100** includes the display portion **110** and the binding portion **130** arranged opposite to each other, and the bending portion **120** connected between the display portion **110** and the binding portion **130**. Wherein the display portion **110** includes the plane display portion **111** and the first arc surface display portion **112** connected between the plane display portion **111** and the bending portion **120**. One end of the bending portion **120** is connected to the first arc surface display portion **112**, and another end of the bending portion **120** is connected to the binding portion **130**. Wherein in the direction of the light-emitting side of the display module **10**, the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is greater than or equal to the height of the horizontal plane where the top surface of the front frame **20** is located.

(17) Specifically, in the embodiment, the front frame **20** includes a bottom frame **21** located on one side of the supporting layer **200** away from the display panel **100** and a first frame **22** located on one side of the bottom frame **21** close to the first arc surface display portion **112**. And the first frame **22** is perpendicular to the bottom frame **21**. In a direction perpendicular to the display panel **100**, a height of a horizontal plane where a top surface of the first frame **22** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located. Preferably, in this embodiment, a distance between the height of the horizontal plane where the top surface of the first frame **22** is located and the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is $H_{sub.1}$.

(18) Please combine FIG. 1 with FIG. 2, it can be understood that compared with the prior art, one end of the bending portion **120** is connected to the flat display portion **110'**, and another end of the bending portion **120** is connected to the binding portion **130**. Therefore, a reserved space must be provided in the front frame **20**, and the bending portion **120** is accommodated in the reserved space. In the direction of the light-emitting side of the display module **10**, the height of the horizontal plane where the top surface of the front frame **20** is located is usually greater (about 2 mm) than the height of the horizontal plane where the top surface of the display portion **110** is located. In the embodiment, the display portion **110** includes the plane display portion **111** and the first arc surface display portion **112** connected between the plane display portion **111** and the bending portion **120**. The one end of the bending portion **120** is connected to the first arc surface display portion **112**, and the another end of the bending portion **120** is connected to the binding portion **130**, so as to reduce the height of the front frame **20** of the existing display device **1**. In the direction of the light-emitting side of the display module **10**, the height of the horizontal plane where the top surface of the first frame **22** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located, so as to eliminate the occurrence of adverse phenomena such as the motion interference caused by the touch between fingers and the edge of the front frame **20** when swiping the screen in the prior art.

(19) Furthermore, in the embodiment, the supporting layer **200** is located on one side of the display portion **110** close to the binding portion **130**. The supporting layer **200** includes the first supporting

sub portion **210** arranged corresponding to the plane display portion **111** and the second supporting sub portion **220** arranged corresponding to the first arc surface display portion **112**, and the shape of the second supporting sub portion **220** is matched with the shape of the first arc surface display portion **112**. Specifically, the first supporting sub portion **210** and the plane display portion **111** are attached with each other, and the two are plane structures with matching shapes. The first supporting sub portion **210** is used to support the plane display portion **111**. The second supporting sub portion **220** and the first arc surface display portion **112** are attached with each other, and they are arcs with matching shapes. The first supporting sub portion **210** is used to support the plane display portion **111**, so as to maintain a shape of the display panel **100**.

(20) In the embodiment, the display panel **100** further includes a second arc surface display portion **113** extending along a direction of the plane display portion **111** away from the first arc surface display portion **112**. The supporting layer **200** includes a third supporting sub portion **230** extending along a direction of the first supporting sub portion **210** away from the second supporting sub portion **220**. The third supporting sub portion **230** is arranged corresponding to the second arc surface display portion **113**, and a shape of the third supporting sub portion **230** is matched with a shape of the second arc surface display portion **113**. The front frame **20** further includes a second frame **23** located on one side of the bottom frame **21** close to the second arc surface display portion **113**, and the second frame **23** is perpendicular to the bottom frame **21**. Wherein in the direction perpendicular to the display panel **100**, a height of a horizontal plane where a top surface of the second frame **23** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located. Preferably, in this embodiment, a distance between the height of the horizontal plane where the top surface of the second frame **23** is located and the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is $H_{sub.2}$.

(21) It can be understood that in this embodiment, the display portion **110** is set to include the plane display portion **111**, the first arc surface display portion **112** connected between the plane display portion **111** and the bending portion **120**, and the second arc surface display portion **113** extending along the direction of the plane display portion **111** away from the first arc surface display portion **112**, making the display panel **100** a hyperboloid structure. Preferably, a radius of curvature of the first arc surface display portion **112** is same as a radius of curvature of the second arc surface display portion **113**, which improves the screen-to-body ratio and aesthetics of the display panel **100**. In addition, the one end of the bending portion **120** is connected to the first arc surface display portion **112**, and the another end of the bending portion **120** is connected to the binding portion **130**. In the direction perpendicular to the display panel **100**, the height of the horizontal plane where the top surface of the first frame **22** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located, and the height of the horizontal plane where the top surface of the second frame **23** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located, so as to eliminate the occurrence of adverse phenomena such as the motion interference caused by the touch between fingers and the edge of the front frame **20** when swiping the screen in the prior art.

(22) Wherein in this embodiment, the supporting layer **200** includes the first supporting sub portion **210** arranged corresponding to the plane display portion **111**, the second supporting sub portion **220** arranged corresponding to the first arc surface display portion **112**, and the third supporting sub portion **230** arranged corresponding to the second arc surface display portion **113**, so as to maintain the shape of the display panel **100**.

(23) In this embodiment, the functional film layer **600** can be attached to a side of the display panel **100** away from the supporting layer **200** through an adhesive layer (not shown in the figure). The functional film layer **600** includes, but is not limited to, a polarizer **610** located on the light-emitting side of the display panel **100**. An orthographic projection of the functional film layer **600**

in the direction perpendicular to the display module **10** covers the plane display portion **111**, the first arc surface display portion **112**, and the second arc surface display portion **113**. The encapsulation structure **700** can be attached to the side of the functional film layer away from the display panel **100** through an adhesive layer (not shown in the figure). The shape of the encapsulation structure **700** is matched with a shape of the functional film layer **600**, and can be attached to the side of the functional film layer **600** away from the display panel **100** through the adhesive layer. The encapsulation structure **700** includes, but is not limited to, a flexible substrate **710** and a flexible cover plate **720** that are successively away from the display panel **100**. The flexible substrate **710** is located between the functional film layer **600** and the flexible cover plate **720**. A material of the flexible substrate **710** includes, but is not limited to, one of cover window (CW) and ultra-thin glass (UTG). The flexible cover plate **720** includes, but is not limited to, a combination of cover plate material protective window (PW) and optically clear adhesive (OCA). Furthermore, a material of the cover plate material protective window includes, but is not limited to, one of polyethylene glycol terephthalate (PET) and colorless polyimide (CPI).

(24) It can be understood that the functional film layer **600** includes the polarizer **610**, and the encapsulation structure **700** includes the flexible substrate **710** and the flexible cover plate **720**, which are only used for illustration. There are no specific restrictions on a film structure of the functional film layer **600** and a film structure of the encapsulation structure **700** in this embodiment. For example, in an embodiment, the encapsulation structure further includes a circulating protective film (CPF) located on one side of the flexible cover plate **720** away from the flexible substrate **710**, and the circulating protective film is attached on one side of the flexible cover plate **720** away from the display panel **100**. When the circulating protective film is scratched during use, customers can go to a maintenance point for replacement to prevent long-term poor appearance caused by scratches. In an embodiment, a depolarizing sheet-**610** (POL less) technology can also be applied to prepare the display module **10**.

(25) It should be noted that in this embodiment, a height of a horizontal plane where a top surface of the flexible cover plate **720** is located is greater than or equal to the height of the horizontal plane where the top surface of the front frame **20** is located.

(26) In this embodiment, the display device **1** further includes a binding component **800** connected to the binding portion **130**. In a direction from the second frame **23** pointing to the first frame **22**, a distance between the binding portion **130** and the bottom frame **21** gradually decreases. Wherein the binding component **800** includes, but is not limited to, a flexible printed circuit (FPC) and an integrated circuit (not shown in figure).

(27) Specifically, the display device **1** further includes a supporting member **500**, and the supporting member **500** is located between the binding portion **130** and the first supporting sub portion **210**. Wherein in the direction from the second frame **23** pointing to the first frame **22**, a thickness of the supporting member **500** gradually increases, so as to increase a distance between the binding component **800** and the bottom frame **21**, and prevent the binding component **800** from directly contacting the bottom frame **21**, which reduces wear of the binding component **800** and improves service life of the display module **10**.

(28) Furthermore, in this embodiment, an orthographic projection of the supporting member **500** on the binding portion **130** is located within a boundary of the binding portion **130**, and an orthographic projection of the supporting member **500** on the first supporting sub portion **210** is located within a boundary of the first supporting sub portion **210**. Because the binding portion **130** and the first supporting sub portion **210** are planar structures, the supporting member **500** can be better born between the binding portion **130** and the first supporting sub portion **210**. When a film layer is bent and attached, the film layers of the display panel **100** will not be deformed and imprinted due to extrusion between the supporting member **500** and a film layer with low hardness or stiffness.

(29) In this embodiment, the display device **1** further includes a heat dissipation layer **400** and the

first backplane **310** both located between the display panel **100** and the supporting layer **200**, and the second backplane **320** located between the binding portion **130** and the supporting member **500**. Wherein the heat dissipation layer **400** is attached to one side of the supporting layer **200** close to the display panel **100**, and a shape of the heat dissipation layer **400** is matched with a shape of the supporting layer **200**. The heat dissipation layer **400** includes, but is not limited to, a metal copper layer, a pressure-sensitive adhesive layer, a white latex layer, a foam rubber layer, and a heat dissipation film layer that are successively away from the supporting layer **200**. The first backplane **310** is attached to one side of the display panel **100** close to the supporting layer **200**, and a shape of the first backplane **310** is matched with a shape of the display panel **100**. The second backplane **320** is attached between the binding portion **130** and the supporting member **500**; and the orthographic projection of the supporting member **500** covers an orthographic projection of the second backplane **320** in the direction perpendicular to the display panel **100**.

(30) Furthermore, in this embodiment, the display module **10** includes at least two non-bending areas D and a bendable area E arranged between adjacent two non-bending areas D. At least a part of the plane display portion **111** is located in the bendable area E, and both the first arc surface display portion **112** and the second arc surface display portion **113** are located in the non-bending areas D. Wherein the first supporting sub portion **210** includes a grid structure **211** located in the bendable area E, so as to improve bending performance of the display module **10** and prevent the film layer from debonding, fracture, and other phenomena during a bending process of the display module **10**.

(31) In an embodiment, please refer to FIG. 3, which is a second schematic structural diagram of the display device provided in the embodiment of the present application.

(32) In this embodiment, a structure of the display module is similar/same as the first structure of the display module provided in the above-mentioned embodiment. Please refer to the description of the display module in the above-mentioned embodiment for details. It will not be repeated here. A difference between the two is only in that:

(33) in this embodiment, the display module **10** is a flexible display module. The display portion **110** includes a rolling portion **110A**, and a non-rolling portion **110B** connected between the rolling portion **110A** and the bending portion **120**. The display device **1** includes a rolling shaft **30** located on one side of the supporting layer **200** away from the display panel **100** and arranged close to the rolling portion **110A**. The rolling portion **110A** is wound on the rolling shaft **30**, and the rolling shaft **30** is used to roll up or unroll the display module **10**. Wherein the non-rolling portion **110B** includes the first arc surface display portion **112**, and the rolling portion **110A** includes the second arc surface display portion **113** extending along a direction of the plane display portion **111** away from the first arc surface display portion **112**. When the display panel **100** is in a unrolled state, the second arc surface display portion **113** and the first arc surface display portion **112** are symmetrically arranged with respect to a horizontal centerline W.sub.1 of the plane display portion **111** in the direction perpendicular to the display panel **100**. It should be noted that this embodiment does not make specific restrictions on a method of winding the rolling portion **110A** on the rolling shaft **30** and a method of the rolling shaft **30** driving the rolling portion **110A** to rotate.

(34) It can be understood that, in this embodiment, the display device **1** includes the rolling shaft **30** located on the side of the supporting layer **200** away from the display panel **100** and arranged close to the rolling portion **110A**, and the rolling portion **110A** is wound on the rolling shaft **30**. When the display panel **100** is in the unrolled state, the second arc surface display portion **113** and the first arc surface display portion **112** are symmetrically arranged with respect to the horizontal centerline W.sub.1 of the plane display portion **111** in the direction perpendicular to the display panel **100**, so that the display panel **100** can be kept as a whole when the display panel **100** is in the unrolled state, which not only can realize a high screen-to-body ratio of an overall screen, but also can improve aesthetics of the display panel **100**.

(35) Furthermore, the first supporting sub portion **210** includes a supporting rolling portion **210A**

arranged corresponding to the rolling portion **110A**. An orthographic projection of the supporting rolling portion **210A** on the display panel **100** overlaps the rolling portion **110A**; and a shape of the supporting rolling portion **210A** is matched with a shape of the rolling portion **110A**, so that the display panel **100** is evenly unrolled, which prevents local warpage of the display panel **100** and makes the display panel **100** more stable in use. Wherein the supporting rolling portion **210A** is wound on the rolling shaft **30** and includes a plurality of hollowing portions (not shown in figure). (36) Specifically, the hollowing portions include a plurality of hollowing holes. This embodiment does not make specific restrictions on shapes of the hollowing holes. Preferably, in this embodiment, the hollowing holes penetrate the supporting rolling portion **210A**; and the hollowing hole is a central symmetrical structure, so as to reduce bending stiffness of the supporting rolling portion **210A** and improve flexibility of the rolling portion **110A** of the display module **10**, so that it can well adapt to and bear rolling or unrolling driven by the rolling shaft **30**. Furthermore, a risk of debonding and fracture of the film layers of the display module **10** is reduced, and an overall yield and reliability of the display module **10** are improved.

(37) Furthermore, in this embodiment, the bottom frame **21** includes a first bottom frame **21A** and a second bottom frame **21B** slidably connected to each other. An orthographic projection of the first bottom frame **21A** in the direction perpendicular to the display module **10** at least partially overlaps with an orthographic projection of the second bottom frame **21B** in the direction perpendicular to the display module **10**. The first frame **22** is located on one side of the first bottom frame **21A** close to the first arc surface display portion **112**. The front frame **20** further includes a second frame **23** located on one side of the second bottom frame **21B** close to the rolling shaft **30**, and the second frame **23** is perpendicular to the second frame **21B**. Wherein in the direction perpendicular to the display panel **100**, a height of a horizontal plane where a top surface of the second frame **23** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located. It should be noted that in this embodiment, there are no specific restrictions on a sliding link mode of the first bottom frame **21A** and the second bottom frame **21B**.

(38) It can be understood that in this embodiment, by setting a sliding link between the first bottom frame **21A** and the second bottom frame **21B**, the orthographic projection of the first bottom frame **21A** in the direction perpendicular to the display module **10** at least partially overlaps the orthographic projection of the second bottom frame **21B** in the direction perpendicular to the display module **10**, so as to prevent influence of the front frame **20** on the rolling or unrolling of the display panel **100**. Moreover, in the direction perpendicular to the display panel **100**, the height of the horizontal plane where the top surface of the first frame **22** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located, and the height of the horizontal plane where the top surface of the second frame **23** is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure **700** is located, so as to eliminate the occurrence of adverse phenomena such as the motion interference caused by the touch between fingers and the edge of the front frame **20** when swiping the screen in the prior art. Preferably, in this embodiment, a distance between the height of the horizontal plane where the top surface of the first frame **22** is located and the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is $H_{sub.1}$, and a distance between the height of the horizontal plane where the top surface of the second frame **23** is located and the height of the horizontal plane where the top surface of the encapsulation structure **700** is located is $H_{sub.2}$.

(39) Furthermore, in this embodiment, the display device **1** further includes a binding component **800** connected to the binding portion **130** and a supporting member **500** located between the binding portion **130** and the first supporting sub portion **210**. Wherein in a direction from the second frame **23** pointing to the first frame **22**, a thickness of the supporting member **500** gradually increases, so as to increase a distance between the binding component **800** and the first bottom

frame 21A, which can prevent the binding component 800 from directly contacting the bottom frame 21, so as to reduce wear of the binding component 800 and improve service life of the display module 10.

(40) A mobile terminal is provided in the embodiment of the present application. The mobile terminal includes a terminal body and the display device described in any one of the above-mentioned embodiments, and the terminal body is combined with the display device as a whole.

(41) It can be understood that the display module has been described in detail in the above-mentioned embodiments, and the description is not repeated here.

(42) In specific applications, the mobile terminal can be a display screen of a device such as smart phones, tablets, laptops, smart bracelets, smart watches, smart glasses, smart helmets, desktop computers, smart TVs, or digital cameras, and even can be applied to electronic devices with flexible display screens.

(43) The display device and the mobile terminal are provided in the present application. The display device includes the display module, and the display module includes the display panel and the supporting layer located at the backlight side of the display panel. The display panel includes the display portion and the binding portion arranged opposite to each other, and the bending portion connected between the display portion and the binding portion. By setting the display portion to include the plane display portion and the first arc surface display portion connected between the plane display portion and the bending portion, the height of the front frame in the existing display device is reduced, so that the orthographic projection of the display portion in the direction perpendicular to the display module covers the bending portion. In the direction of the light-emitting side of the display module, the height of the horizontal plane where the top surface of the display portion is located is greater than or equal to the height of the horizontal plane where the top surface of the front frame is located, so as to eliminate the occurrence of adverse phenomena such as the motion interference caused by the touch between fingers and the edge of the front frame when swiping the screen in the prior art. Moreover, by setting the supporting layer to include the first supporting sub portion corresponding to the plane display portion and the second supporting sub portion corresponding to the first arc surface display portion, the orthographic projection of the first supporting sub portion on the display panel overlaps the plane display portion, and the orthographic projection of the second supporting sub portion on the display panel overlaps the first arc surface display portion. The shape of the second supporting sub portion is matched with the shape of the first arc surface display portion, so as to support the display panel.

(44) In summary, although the present application has been disclosed as above with preferred embodiments, the above-mentioned preferred embodiments are not intended to limit the invention. Ordinary technicians in the art can make various changes and refinements without departing from the spirit and scope of the invention. Therefore, the scope of protection of the present application is subject to the scope defined in the claims.

Claims

1. A display device, comprising a front frame and a display module located in the front frame, and the display module comprising: a display panel comprising a display portion and a binding portion arranged opposite to each other, and a bending portion connected between the display portion and the binding portion; and the display portion comprising a plane display portion and a first arc surface display portion connected between the plane display portion and the bending portion; an encapsulation structure arranged on a light-emitting side of the display panel; and a supporting layer located on one side of the display portion close to the binding portion, and the supporting layer comprising a first supporting sub portion arranged corresponding to the plane display portion and a second supporting sub portion arranged corresponding to the first arc surface display portion; and a shape of the second supporting sub portion being matched with a shape of the first arc surface

display portion; wherein in a direction of the light-emitting side of the display module, a height of a horizontal plane where a top surface of the encapsulation structure is located is greater than or equal to a height of a horizontal plane where a top surface of the front frame is located.

2. The display device as claimed in claim 1, wherein the front frame comprises a bottom frame located on one side of the supporting layer away from the display panel and a first frame located on one side of the bottom frame close to the first arc surface display portion, and the first frame is perpendicular to the bottom frame; and in a direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the first frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

3. The display device as claimed in claim 2, wherein the display portion comprises a second arc surface display portion extending along a direction of the plane display portion away from the first arc surface display portion; the supporting layer comprises a third supporting sub portion extending along a direction of the first supporting sub portion away from the second supporting sub portion; the third supporting sub portion is arranged opposite to the second arc surface display portion, and a shape of the third supporting sub portion is matched with a shape of the second arc surface display portion; the front frame further comprises a second frame located on one side of the bottom frame close to the second arc surface display portion, and the second frame is perpendicular to the bottom frame; and wherein in the direction perpendicular to the display panel, a height of a horizontal surface where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

4. The display device as claimed in claim 3, wherein a radius of curvature of the first arc surface display portion is same as a radius of curvature of the second arc surface display portion.

5. The display device as claimed in claim 3, wherein the display device further comprises a binding component connected to the binding portion, and in a direction from the second frame pointing to the first frame, a distance between the binding portion and the bottom frame gradually decreases.

6. The display device as claimed in claim 5, wherein the display device comprises a supporting member, and the supporting member is located between the binding portion and the first supporting sub portion; and wherein in the direction from the second frame pointing to the first frame, a thickness of the supporting member gradually increases.

7. The display device as claimed in claim 3, wherein the display module comprises at least two non-bending areas and a bendable area arranged between adjacent two of the non-bending areas; at least a part of the plane display portion is located in the bendable area, and both the first arc surface display portion and the second arc surface display portion are located in the non-bending areas; and wherein the first supporting sub portion comprises a grid structure located in the bendable area.

8. The display device as claimed in claim 7, wherein a material of the supporting layer is metal.

9. The display device as claimed in claim 2, wherein the display portion comprises a rolling portion and a non-rolling portion connected between the rolling portion and the bending portion; the display device comprises a rolling shaft located on the one side of the supporting layer away from the display panel and arranged close to the rolling portion; and the rolling portion is wound on the rolling shaft, and the rolling shaft is used to roll up or unroll the display module; and wherein the non-rolling portion comprises the first arc surface display portion, and the rolling portion comprises a second arc surface display portion along a direction of the plane display portion away from the first arc surface display portion; when the display panel is in a unrolled state, in the direction perpendicular to the display panel, the second arc surface display portion and the first arc surface display portion are symmetrically arranged with respect to a horizontal centerline of the plane display portion.

10. The display device as claimed in claim 9, wherein the first supporting sub portion comprises a supporting rolling portion arranged corresponding to the rolling portion, and a shape of the supporting rolling portion is matched with a shape of the rolling portion; wherein the supporting rolling portion is wound on the rolling shaft, and the supporting rolling portion comprises a

plurality of hollowing portions; and wherein the front frame further comprises a second frame located on one side of the bottom frame close to the rolling portion, and the second frame is perpendicular to the bottom frame; and in the direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

11. The display device as claimed in claim 10, wherein the display device further comprises a binding component connected to the binding portion and a supporting member located between the binding portion and the first supporting sub portion; and wherein in a direction from the second frame pointing to the first frame, a thickness of the supporting member gradually increases.

12. The display device as claimed in claim 1, wherein the encapsulation structure comprises a flexible substrate and a flexible cover plate arranged in layers, and the flexible substrate is located between the display panel and the flexible cover plate; and wherein a material of the flexible substrate is one of cover window and ultra-thin glass; and the flexible cover plate is a combination of a cover plate material protective window and an optically clear adhesive, and a material of the cover plate material protective window is one of polyethylene glycol terephthalate and colorless polyimide.

13. A mobile terminal, comprising a terminal body and a display device, the terminal body and the display device being integrated, and the display device comprising a front frame and a display module located in the front frame, and the display module comprising: a display panel, the display panel comprising a display portion and a binding portion arranged opposite to each other, and a bending portion connected between the display portion and the binding portion; and the display portion comprising a plane display portion and a first arc surface display portion connected between the plane display portion and the bending portion; an encapsulation structure arranged on a light-emitting side of the display panel; and a supporting layer located on one side of the display portion close to the binding portion, and the supporting layer comprising a first supporting sub portion arranged corresponding to the plane display portion and a second supporting sub portion arranged corresponding to the first arc surface display portion; and a shape of the second supporting sub portion being matched with a shape of the first arc surface display portion; wherein in a direction of the light-emitting side of the display module, a height of a horizontal plane where a top surface of the encapsulation structure is located is greater than or equal to a height of a horizontal plane where a top surface of the front frame is located.

14. The mobile terminal as claimed in claim 13, wherein the front frame comprises a bottom frame located on one side of the supporting layer away from the display panel and a first frame located on one side of the bottom frame close to the first arc surface display portion, and the first frame is perpendicular to the bottom frame; and in a direction perpendicular to the display panel, a height of a horizontal plane where a top surface of the first frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

15. The mobile terminal as claimed in claim 14, wherein the display portion comprises a second arc surface display portion extending along a direction of the plane display portion away from the first arc surface display portion; the supporting layer comprises a third supporting sub portion extending along a direction of the first supporting sub portion away from the second supporting sub portion; the third supporting sub portion is arranged opposite to the second arc surface display portion, and a shape of the third supporting sub portion is matched with a shape of the second arc surface display portion; the front frame further comprises a second frame located on one side of the bottom frame close to the second arc surface display portion, and the second frame is perpendicular to the bottom frame; and wherein in the direction perpendicular to the display panel, a height of a horizontal surface where a top surface of the second frame is located is less than or equal to the height of the horizontal plane where the top surface of the encapsulation structure is located.

16. The mobile terminal as claimed in claim 15, wherein a radius of curvature of the first arc surface display portion is same as a radius of curvature of the second arc surface display portion.

17. The mobile terminal as claimed in claim 15, wherein the display device further comprises a binding component connected to the binding portion, and in a direction from the second frame pointing to the first frame, a distance between the binding portion and the bottom frame gradually decreases.

18. The mobile terminal as claimed in claim 17, wherein the display device comprises a supporting member, and the supporting member is located between the binding portion and the first supporting sub portion; and wherein in the direction from the second frame pointing to the first frame, a thickness of the supporting member gradually increases.

19. The mobile terminal as claimed in claim 15, wherein the display module comprises at least two non-bending areas and a bendable area arranged between adjacent two of the non-bending areas; at least a part of the plane display portion is located in the bendable area, and both the first arc surface display portion and the second arc surface display portion are located in the non-bending areas; and wherein the first supporting sub portion comprises a grid structure located in the bendable area.

20. The mobile terminal as claimed in claim 19, wherein a material of the supporting layer is metal.
